

SEMWISE: A national soil health scoring framework for agricultural systems

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ABSTRACT

Soil health is an essential part of agricultural productivity, food security, and ecological sustainability. In the U.S., national, state, and local programs invest millions of dollars each year to support soil health and climate-smart agricultural practices. While several methods for measuring soil health are available, the ability to interpret soil health metrics across regions and agricultural ecosystems remains challenging. There is not yet a nationally scaled framework that incorporates important soil health indicators, soil health principles, and influential variables like climate and soil texture in one model. Here, we introduce SEMWISE, a Structural Equation Model for Well-Informed Soil Evaluation, that was developed from 600 samples across 28 U.S. states representing 52 different crops and grazing lands. SEMWISE incorporates seven soil health indicators (aggregate stability, water capacity, soil organic matter, active carbon, autoclaved-citrate extractable protein, respiration, and quantitative-PCR) into individual and collective ratings that adjust for regional differences in soil clay content and climate, allowing for a widely applicable framework that can detect management responses at local and national scales. Furthermore, we developed the first soil health management index (SHMI) that evaluates the agricultural system using the soil health principles to minimize soil disturbance, increase plant diversity, and provide continuous soil cover and living roots. The SEMWISE and SHMI frameworks were robust, sensitive, and scalable for detection of soil health differences at national and local levels.

1. Introduction

Soil health is a critical component of agricultural productivity and food security. Better understandings of the physical, chemical, and biological components of soil health has led to knowledge about its ability to provide essential nutrients, water, and other ecosystem services that support plant growth and resilience to biotic and abiotic stresses (Bagnall et al., 2021; Brevik et al., 2020; Costanza et al., 1997; Moebius-Clune et al., 2016). As defined, soil health is an integrated concept that reflects the “capacity of soil to function, within land use and ecosystem boundaries, to sustain biological productivity, maintain environmental quality, and promote plant material and human health” (Doran and Parkin, 1994). Common management practices are based on the soil health principles defined by the Natural Resources Conservation Service (NRCS), including increased plant and animal diversity, less soil

disturbance, a continuous presence of plant growth/roots in the soil, and increased soil cover (Karlen et al., 2021). In the U.S. alone, a resurgence for the importance of soil health is illustrated by huge national public and public-private investments to promote practices to develop soil health management systems. It has been estimated that up to \$122 million per year of the total U.S. Department of Agriculture budget is allocated to practices that improve soil health, transition to ecologically-based management, and expand adaptive strategies (Basche et al., 2020).

The concept of soil health is not new and research efforts to quantify and monitor how agronomic practices affect soil function have been occurring for decades (Karlen et al., 2019). Given the multitude of possible soil health functions and metrics available (Bagnall et al., 2023; Karlen et al., 2021, 2019; Stott, 2019), a definitive suite of soil health indicators remains a high priority research gap (Bünemann et al., 2018;

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Stewart et al., 2018) and efforts continue to make advances to achieve consensus and consistency (Bagnall et al., 2023; Nunes et al., 2021). While massive sample collection undertakings have been accomplished, consistently ranking soil health has been a difficult task, primarily due to inherent (e.g., mineralogy and texture) and climatic differences among soils (Karlen et al., 2021). Importantly, methods to integrate the many soil health indicators into an overall interpretation of soil health faces statistical challenges. Challenges arise because most statistical approaches require weighting of indicators in a subjective way, preventing quantifiable validation and consistency between measurements (Maaz et al., 2023).

Finding a consistent way to interpret soil health measurements across many climatic regions is an essential step towards completing comprehensive and inclusive studies of soil health across the world (Brevik et al., 2020; Veum et al., 2021). The development of a framework for national (and/or international) assessments of soil management practices has been emphasized as an important soil health research and technology transfer activity (Karlen et al., 2021). The ability to interpret soil health at regional, national, and international scales can help identify areas where soil health is deficient and guide management practices towards improvement, thus promoting sustainable agricultural practices and ecosystem services (Bagnall et al., 2021; Brevik et al., 2020; Costanza et al., 1997; Moebius-Clune et al., 2016; Veum et al., 2021).

Several new and more analytical ways of interpreting soil health based on indicators (Ball et al., 2017; Bünemann et al., 2018; Emmet-Booth et al., 2016; Hussain et al., 2022; Karlen et al., 2021; Rinot et al., 2019) have been surfacing since the 1990s across many countries, and some techniques have become popular. In the U.S., two of the more widely recognized methods include the Soil Management Assessment Framework (SMAF) (Wienhold et al., 2009) and the Comprehensive Assessment of Soil Health (CASH) (Moebius-Clune et al., 2016). However, many of these analytical-based methods, including SMAF and CASH, used relatively small data sets for their development and are not calibrated nationwide or globally (Nunes et al., 2021). Another more recent approach, the Soil Health Assessment Protocol and Evaluation (SHAPE) tool (Nunes et al., 2021), has a much higher sample size (over 14 K) and includes multiple climates across the U.S. While the SHAPE framework additionally includes multiple indicators (soil organic carbon (SOC), beta-glucosidase, aggregate stability, autoclaved citrate-extractable (ACE) protein, active C, and respiration), to our knowledge the framework has only been fully applied to SOC, with expansions currently in progress. The CASH approach was updated in 2017 to expand interpretations from the original focus on north-east soils to include mid-Atlantic and Midwest regions (Fine et al., 2017), and recent efforts by the Soil Health Institute and partners have begun to interpret individual soil health indicators at the continental scale (Liptzin et al., 2022; Rieke et al., 2022a; Rieke et al., 2022b). However, an integration of all indicators, adjusted and appropriately scaled at a national level, and linked to the soil health principles, has yet to be achieved. In part, this is due to differing soil types between regions, the complex interactions between land use, and diverse management practices that contribute to the principles.

Structural equation modeling (SEM), “a set of statistical techniques used to measure and analyze the relationships of observed and latent variables” by examining “linear causal relationships among variables, while simultaneously accounting for measurement error” (Beran and Violato, 2010), may be useful for developing such a tool. Some researchers have been exploring SEM to score soil health. In Wade et al. (2020), a multilevel SEM was used to show that biologically healthier soils produce greater corn yields per unit of fertilizer across the central and eastern Corn Belt of the Midwestern U.S. In that study, a soil health latent variable was defined using permanganate oxidizable carbon, autoclave citrate-extractable protein, and soil organic content. In Maaz et al. (2023), a scoring function using SEM was developed using soils collected from 50 sites across six islands within the Hawaiian

archipelago. The function defined a soil health latent variable using nine indicators and adjusted scores by inherent soil properties as well as land use (Maaz et al., 2023).

SEM is beneficial because it offers simultaneous quantification of effects across scales, balances type I and type II errors, and has accurate model predictions across similar sites (Wade et al., 2020). When combined with factor analysis, SEM is particularly robust in describing soil health traits across contexts (Wade et al., 2022), and it provides a flexible framework for analyzing complex relationships among multiple variables, allowing researchers to validate theories using empirical models (Beran and Violato, 2010). More specifically, SEM can test the direct and indirect effects (Fan et al., 2016) of texture, climate, and soil health indicators on one soil health latent variable in a single model. However, before SEM tools for soil health interpretation are developed, it is important to validate its accuracy across different climatic regions and land uses.

Connecting interpretations of soil health to management practices is crucial for effective agricultural practices. While it is important to assess and interpret soil health, it is equally essential to establish a connection between these interpretations and the management practices employed by farmers. Soil health management is emerging as an effective natural climate solution with support from many stakeholders (Basche et al., 2020). Despite the increasing popularity of soil health management systems, inconsistencies in interpreting soil health often arise from variations in defining the relationships between soil health indicators and individual management practices. To address this issue, it is imperative to establish a consistent and standardized framework that relates management practices at the system level to soil health, benefiting both farming practices and research endeavors, ultimately contributing to sustainable farming and the long-term viability of agricultural systems.

The objectives of this study were to: (1) utilize a national soil health survey using samples collected from multiple states and agricultural ecosystems to develop a flexible and scalable soil health interpretative platform; and (2) develop a soil health management scoring index that represents the principles of soil health management. To achieve these goals, we leveraged soil health indicator data collected from over 600 samples across 28 U.S. states representing 52 different crops and grazing lands and developed and tested a new Structural Equation Model for Well-Informed Soil Evaluation (SEMWISE). Furthermore, we developed a semi-quantitative, integrated soil health management index (SHMI) that aligned with the soil health principles and explored its relationship with the overall SEMWISE soil health score.

2. Methods

2.1. Sample collection

A total of 632 soil samples (0–15 cm) were collected between May 2016 and July 2018 from 26 states across the U.S. with 521 from cropland systems and 111 from rangeland (also includes hay land and pasture). Sample collection was coordinated by members of the NRCS Soil Health Division as part of a national soil health initiative aimed to increase awareness and engage farmers and local and regional conservation partners. The sites consisted of a mixture of research plots with replicated samples and non-research sites from individual, private farms and ranches (Table S1). Samples were split into land use categories of annual cropland (N = 412), perennial cropland (N = 109), and rangeland (N = 111). Köppen-Geiger climate zones (Kottek et al., 2006), which account for temperature and precipitation, were also identified for each sampling location using the *kgc* package (Bryant et al., 2017) in R (R Core Team, 2017). The climate zones represented in these data and their corresponding codes include cold semi-arid (steppe) (BSk), humid subtropical (Cfa), hot-summer Mediterranean (Csa), warm-summer Mediterranean (Csb), hot-summer humid continental (Dfa), warm-summer humid continental (Dfb), subarctic climate (Dfc), and warm,

dry-summer continental (Dsb).

2.2. Soil health and molecular assays

Samples were submitted for the Comprehensive Assessment of Soil Health (CASH) at Cornell University Soil Health Lab. This service was used because its convenience and standardization would make the adoption of this model easier for other researchers and farmers. We used results from two physical indicators (wet aggregate stability [Agg Stab], available water capacity [Water Capacity]), four biological indicators (soil organic matter [SOM], active carbon [Active C], autoclaved-citrate extractable protein [ACE], and soil respiration [Resp]), and soil pH and a suite of macro- and micro-nutrients for the chemical indicators. Details of all protocols are provided by [Schindelbeck and Moebius-Clune \(2016\)](#). Briefly, Agg Stab was determined using a custom-designed rainfall simulator which produces an unstable, 'slaked' fraction and a water stable aggregate fraction. Soil available water capacity was determined from measuring the difference between field capacity and permanent wilting point using pressure plates. Soil organic matter was determined using the loss-on-ignition method and calculated according to the equation reported by [Nelson and Sommers \(1996\)](#). Active C was measured following the potassium permanganate oxidation method as described by [Weil et al. \(2003\)](#). Resp was estimated by measuring the amount of CO₂ released in a 4-day incubation following rewetting of an air-dried sample. ACE was determined by extracting proteins in a sodium citrate solution neutralized with citric acid and exposed to heat using an autoclave ([Hurisso et al., 2018](#)). The proteins were extracted and quantified colorimetrically. Soil pH was measured in a 2:1 water:soil ratio using a Lignin pH robot (Albuquerque, NM). Extractable phosphorus, potassium, magnesium, iron, manganese, and zinc were determined using a Modified Morgan extractant and measured by inductively coupled plasma analysis. Although not a discrete indicator of soil health, soil clay, silt, and sand contents are important for accurate interpretations and were determined using the rapid soil particle-size method as described by [Kettler et al. \(2001\)](#).

Soil subsamples were frozen and shipped to the U.S. Department of Agriculture, Agricultural Research Service, Fort Collins, CO for molecular characterization of the microbial community as part of a larger research effort seeking to incorporate microbial data into interpreting soil health. For the purpose of this study, quantitative PCR (qPCR) was used to serve as a measure of overall bacterial abundance ([Manter et al., 2021](#)). DNA was extracted from 0.25 g soil samples using the Qiagen DNeasy Powersoil Pro Kit (Qiagen, Germantown, MD) and carried out using a fully automated Qiagen QIAcube robot. DNA quality was assessed using a Nanodrop 1000 (Thermo Scientific, Waltham, MA) and quantified fluorometrically with the Invitrogen dsDNA HS Assay Kit on a Qubit 2.0 (Life Technologies, Carlsbad, CA). Extracted DNA was amplified in triplicate using the 341f/806rb primers that target the V3-V4 region of the rRNA gene as previously described ([DiLegge et al., 2022](#)). The number of 16S rRNA gene copies was determined using external standard curves generated by serial dilution of purified *Pseudomonas putida* KT2440 gDNA. Copy numbers were normalized to the grams of dry soil extracted and used as a proxy of bacterial abundance.

2.3. Data processing

All data processing were performed in R ([R Core Team, 2017](#)) using tidyverse ([Wickham et al., 2019](#)), base R, and/or the mentioned packages. After climate zones were identified (see [Section 2.1](#) Sample collection), samples within the Dsb climate zone were removed due to the low sample size ($n = 8$). An "acidity" metric was calculated by reverse scoring soil pH. Each soil health indicator was checked for normality using a Shapiro-Wilk test ([Shapiro and Wilk, 1965](#)) and normalized using either a Box-Cox transformation ([James et al., 1996](#)) or a log₁₀ transformation. For each indicator, the transformation method was chosen based on whichever gave the lowest p-value after a

Shapiro-Wilk test. Although normality was improved for all indicators, the Shapiro-Wilk tests still suggested that the data were not normally distributed ($p < 0.05$). Outliers were identified using the outlier function in the psych package ([Yuan and Zhong, 2008](#)). Eight outliers were identified and removed from the data set, leaving 616 samples remaining for constructing the structural equation model.

2.4. SEMWISE: Structural Equation Modeling for Well-Informed Soil Evaluation

A structural equation model was constructed based on the approach as described in [Maaz et al. \(2023\)](#). In our model, each soil health indicator was adjusted by clay content for each climate zone to account for the strong effects of these inherent properties as discussed above. Models were constructed using the sem function in the lavaan package ([Rosseel, 2012](#)). Before modeling, the transformed data were centered and scaled using the scale function in base R. Climate zones were split into individual columns so that a binary could indicate the climate zone for each sample (e.g., column "Csa" had values of zero or one for each sample, where one indicated that sample was collected within climate zone Csa).

Analyses using Cattell's scree test and maximum-likelihood factor analysis ($\chi^2 = 1212$, p -value < 0.001) did not support splitting the indicators into additional underlying factors. Therefore, all indicators contributed individually to a single first-order soil health (SH) latent variable in developing our SEM model. Additionally, soil acidity and micro- and macro-nutrient concentrations were not significant contributors to the SH latent variable ($p > 0.100$) and were removed from the model. We followed the clustered SEM model approach outlined by [Maaz et al. \(2023\)](#) to account for high intraclass correlations due to the repeated sampling within the research sites. Additional improvement of the clustered SEM model was conducted by adding covariances with modification indices > 10 to the SEM (Table S2). The final clustered SEM with a single soil health latent variable (SEMWISE) predicted from the seven indicators (SOM, Active C, Resp, qPCR, Agg Stab, ACE, and Water Capacity) resulted in fit statistics that were well within recommended cutoffs (e.g., robust comparative fit index (CFI) = 0.999, robust root mean square error of approximation (RMSEA) = 0.028, standardized root mean residual (SRMR) = 0.003, Tables S2–S4) ([Fabrigar et al., 1999](#); [Hooper et al., 2008](#)).

The SH latent variable was converted to a soil health rating using an empirical cumulative distribution function similar to that employed in other soil health ratings ([Amsili et al., 2021](#)). SEMWISE was also used to generate adjusted indicator values by removing the effects of climate zone and clay. Briefly, for each sample, indicator values were predicted based on climate zone and clay contents, using the lavPredictY function in lavaan, and subtracted from each observation. Indicators were then back-transformed to express the adjusted values in their original units (e.g., SEMWISE adjusted). Adjusted values were also converted into individual indicator ratings using the empirical cumulative distribution function described above.

2.5. Calculating the soil health management index and connecting SEMWISE to management

Although the initial sampling effort and metadata collected was not conducted with the intent to develop a management index, we strived to develop a mechanism to evaluate soil health at the system level rather than to compare independent practices (e.g., cover crop vs. no cover crop). To achieve this goal and to connect SEMWISE to management, a semi-quantitative soil health management index (SHMI) was developed by translating cropping system metadata into one of three independent soil health indices. The three indices are the biodiversity index, the disturbance index, and the root/cover index and were designed to align with the soil health principles of (1) using plant diversity to increase diversity in the soil (i.e., the richness index), (2) minimize soil disturbance (i.e., the disturbance index), and (3) maximize the duration of

living roots (i.e., the roots index). The soil health principle of keeping the soil covered was not able to be a discrete index due to a lack of metadata regarding the actual coverage of soil from vegetation or mulch. However, we consider the roots index to capture a large component of this principle given the emphasis on cover cropping and/or perennial-based systems in the metadata.

The richness index was calculated by estimating the average annual plant functional richness over a 2–3-year period. Plants were assigned to one of seven functional categories: warm- and cool-season legumes, warm- and cool-season grasses, warm- and cool-season broadleaf species, and brassicas (Table S5). Among other potential functions, these broad categories were used to evaluate a potential mixture of plants grown over a crop-rotation cycle with different rooting structures (e.g., grasses vs. brassicas), plant growth phases (warm-season vs. cool-season), and the ability to supply or reallocate nutrients (e.g., legumes and deep-rooted grasses). The number of different functional groups in a calendar year, averaged over a 2–3-year period, was calculated. The number of years of metadata provided by the farmers is the number of years used in the average annual functional richness calculation.

The roots index was assumed to represent a combination of soil cover and presence and duration of roots. Typical planting and harvesting dates for each cash crop according to the National Agricultural Statistics Service (NASS) were used (Nass, 2010). Cover crops were assumed to be planted and terminated within appropriate date ranges for each location. When planting and harvesting/termination dates were provided by the farmers, those were used instead. Theoretical root presence was calculated by summing the number of days between all planting and harvest/termination dates for the entire 1–3-year period and converted to a percentage.

To calculate the disturbance index, the tillage type provided by the farmers was used. As detailed information related to the frequency and depth of each physical disturbance was not provided, only the category of tillage could be used. Categories for disturbance included none (e.g., for rangeland), no till, no till/direct seed, strip till, zone till, vertical till, chisel plow, cutting disc, shallow till, conventional till, disc plow, and moldboard plow.

2.6. Statistical analysis

All data manipulations, statistical analyses, and graphing were performed in R (R Core Team, 2017). Data manipulations were performed using tidyverse (Wickham et al., 2019). All statistical comparisons between categorical groups were made in R using the compare_means function in the ggpubr package (Kassambara, 2023) with the non-parametric Wilcoxon rank sum test (Rey and Neuhauser, 2011) and 'fdr' adjustment to minimize false positives. Significance was determined using $p < 0.05$. All regressions were calculated using the lm function in the stats package (Chambers, 1992; Wilkinson and Rogers, 1973) in base R. Graphs were generated in Microsoft PowerPoint or in R using the following packages: ggplot2 (Wickham, 2016), patchwork (Pedersen, 2019), usmap, and sp (Bivand et al., 2013; Pebesma and Bivand, 2005).

3. Results and discussion

3.1. Integrating clay, climate, and soil health indicators in a national soil health comparison model

The first step in the development of SEMWISE, a Structural Equation Model for Well-Informed Soil Evaluation, was to adjust each soil health indicator for the interactive effect of soil texture and climate. The influence of soil texture on soil biological and physical properties is well established, affecting soil microbial communities (Xia et al., 2020), soil carbon and organic matter dynamics (Plante et al., 2006), aggregate stability (Rivera and Bonilla, 2020), and soil health (Amsili et al., 2021). Climate, one of the critical soil forming factors, as proposed by Jenny

(Jenny, 1961), is an equally important factor affecting soil interpretations and thresholds (Nunes et al., 2021). Our study was no exception with clay content and climate zone significantly influencing the soil health indicators (e.g., Fig. 1). To account for these influences, regressions between the indicator values and clay content for each climate zone were included in SEMWISE. Before adjustment, regressions between SOM and clay ($p < 0.001$, $F = 551.8$) and SOM and climate zone ($p < 0.001$, $F = 144.0$) were significant (Fig. 1). After adjustment, there was no significant interaction between SOM and clay or climate zone ($p = 0.949$ and $p = 0.937$, respectively, Fig. 1). This was the case for all seven soil health indicators; their relationships with clay and climate zone were significant before adjustment and non-significant after adjustment (Table S6).

For the final SEMWISE, all seven soil health indicators were positively related to a single soil health latent variable (Figs. 2, S1). Other SEM models for soil health have revealed multiple underlying latent factors where the authors have assigned functional explanations. For example, three factors were found to provide the strongest fit to the data used to evaluate legacy land use impacts in Hawaiian soil that cut across the more traditional biological, chemical, and physical classes (Maaz et al., 2023). A mixture of biological and physical soil health indicators were grouped in another SEM-directed evaluation of soil health on corn yields (Wade et al., 2020), further supporting the concept that these variables are not as clearly delineated as initially thought (Lehmann et al., 2020). Our final SEMWISE resulted in a single soil health latent variable without further aggregation of indicators into secondary latent variables. This may reflect the orthogonality of the indicators chosen for the CASH approach following nearly two decades of research (Fine et al., 2017). Additionally, in comparison to the Hawaiian-focus by Maaz et al. (2023) or the state-level case studies in Wade et al. (2020), our data set represents a much broader suite of samples from highly diverse agroecosystems and management practices across the U.S., which may decouple localized correlations between indicators. As highlighted by Maaz et al. (2023), regardless of how variables are grouped or not, SEM has the advantage of allowing the statistics to assign appropriate weightings to the indicators or aggregated factors rather than the user assigning equal (or other subjective) weightings.

Based on the SEMWISE standardized coefficients (Table S4), Active C had the strongest influence on the soil health latent variable with ACE and SOM the next two most significant contributors based on their standardized coefficients of 0.573, 0.497, and 0.449, respectively. Active C is hypothesized to be the portion of SOM that is readily available as food and energy for the soil microbial community (Moebius-Clune et al., 2016). In addition to the reported positive association of Active C with SOM, Agg Stab, and Resp, it was recently found to be an indicator of corn productivity in Missouri (Svedin et al., 2023), thus linking this indicator to an agronomic outcome. ACE is generally positively associated with increased soil health because organically bound N influences the soil ability to store and mineralize N, thus making N available to plants during the growing season (Geisseler et al., 2019; Hurisso et al., 2018). Additionally, ACE is associated with soil aggregation, water capacity, water movement (Moebius-Clune et al., 2016), and crop yield (Hurisso et al., 2018; Svedin et al., 2023). SEMWISE incorporates each of these indicators into one model, allowing for comprehensive scoring that is informed by indicators known to respond to soil management.

To show the effect that the confounding variables (e.g., clay content and climate) had on the detection of management practices, we examined the distribution of raw and adjusted SOM values in different cover crop types (annual cropland without cover, annual cropland with cover, and perennial) (Fig. 3). Before adjustment, there were no significant differences in SOM values between cover crop types, and perennial cropland/rangeland had lower SOM values than annual croplands with and without cover crops (Fig. 3A). Adjusted SOM values were significantly different between all three cover crop types (Fig. 3B), and perennial cropland had the highest SOM values (Fig. 3B). Clay content

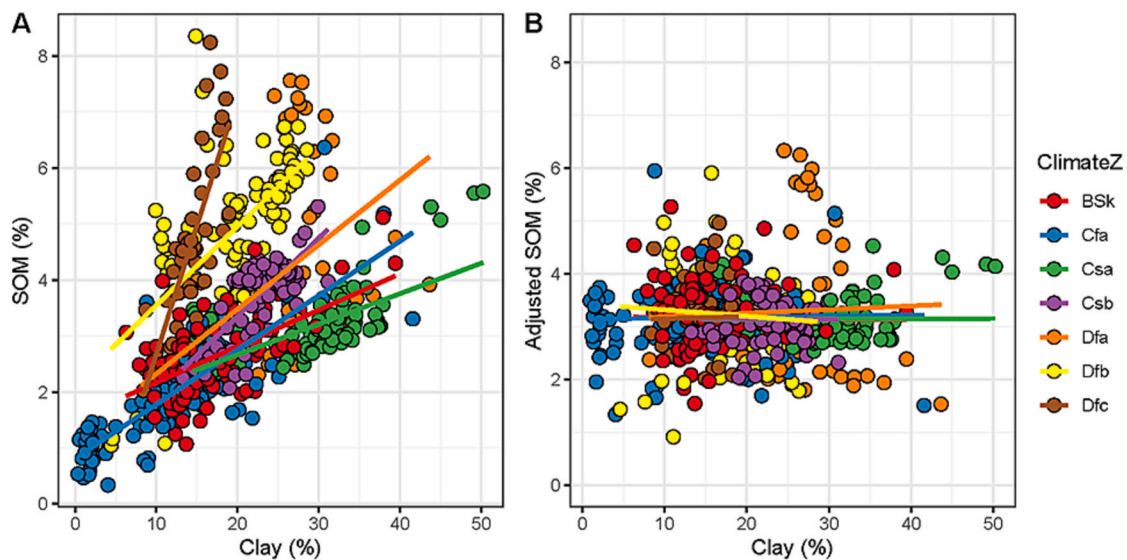


Fig. 1. Relationship between SOM and clay within Köppen climates zones before (A) and after (B) adjustment.

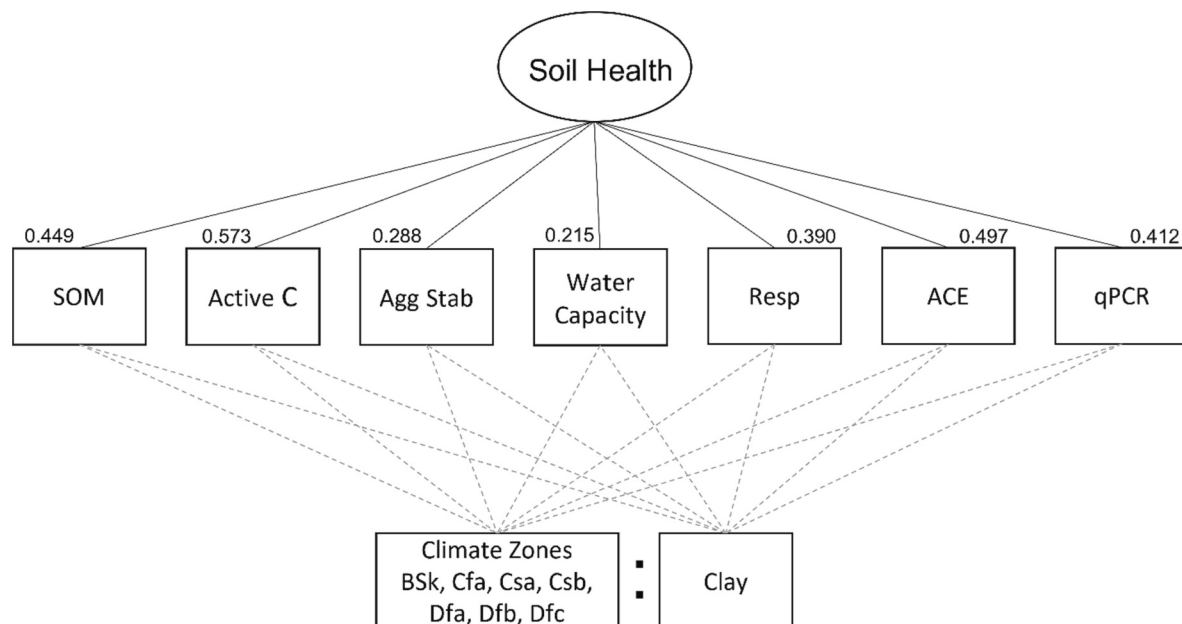


Fig. 2. Final clustered SEMWISE model structure with robust standard errors accounting for sites with high intraclass correlations. Soil health is a latent variable measured by the seven indicators connected by solid lines. Climate zones were not combined into a latent variable (i.e., each indicator was regressed against each zone in SEMWISE), but they are combined in this figure for the sake of simplicity in visualization. Clay content was also incorporated, including interaction terms between clay and each climate zone (as shown by the colon). Values next to the indicators are the standardized regression coefficients for each variable.

and climate are important considerations when deciding on ways to improve soil health, but they can mask the effect of management. Since SEMWISE includes clay, climate, and their interaction in the model structure, the model reduces this masking effect and enables better detection of soil responses to management. Therefore, SEMWISE is robust to scoring and interpreting soil health while adjusting for inherent soil properties.

3.2. Scaling SEMWISE for research to a national scale

To further determine the viability of SEMWISE, it was compared to the overall soil health score reported by CASH. CASH uses cumulative normal distribution functions to standardize the data, ultimately deriving interpretative scores between 0 and 100 (Amsili et al., 2021).

For most soil health indicators, separate scoring curves are used for different categorical textural groupings (e.g., coarse, medium, and fine). Despite improvements to the CASH scoring functions with additional regional assessments (Fine et al., 2017), soil health ratings between climate zones were significantly different in almost all comparisons (Fig. 4A, $p < 0.05$). In contrast, no significant differences between climate zones were observed using the SEMWISE ratings (Fig. 4B, $p < 0.05$). This indicates that SEMWISE does not show as much of a climate effect, and it is more robust to predicting varied levels of soil health within the same climate zone. This is likely due to the broader sample set collected in this study compared to CASH (Moebius-Clune et al., 2016). Furthermore, ratings within climate zones were more varied for SEMWISE than CASH (Fig. 4A–C) indicating the continuous nature of SEMWISE ratings compared to the more narrowly distributed CASH ratings.

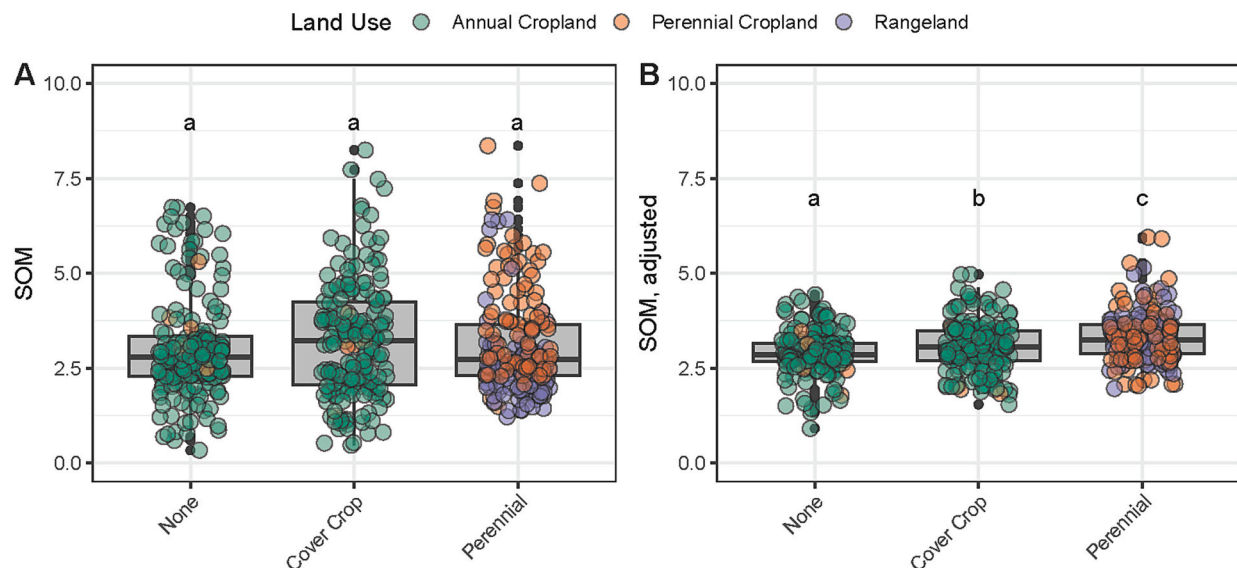


Fig. 3. Unadjusted (A) versus adjusted (B) SOM values between cover crop types. Points represent the individual values for each soil sample and are colored by annual cropland (green), perennial cropland (orange), and rangeland (purple). Statistical comparisons were made using a Wilcoxon rank sum test (significance threshold of $p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

For example, within the Csb climate zone, CASH ratings ranged from 48 to 78, whereas SEMWISE ratings ranged from 4 to 90 (Fig. 4C). Among individual indicators, this distinction was particularly noticeable when comparing SOM content with CASH and SEMWISE SOM ratings within coarse, medium, and fine textures (Fig. S2), showing that SEMWISE allows for a more continuous overlap of scoring between textures while CASH retains a discrete nature. When the CASH scoring curves are updated to include more regions, it would be valuable to determine whether it allows for more varied soil health scores within climate zones.

The overarching goal of SEMWISE was to develop a soil health score that was scalable and sensitive to individual management practices and soil health management systems. Cover cropping is promoted as a key soil health practice with numerous studies reporting on the benefits to water quality (Blanco-Canqui, 2018), economic stability (Bowman et al., 2022), and climate change mitigation (Moore et al., 2023). We tested the effect of cover cropping by grouping the data into annual croplands with and without cover cropping and one group that contained all perennial cropland and rangeland samples. Averaged nationally, the SEMWISE soil health scores were lower in the annual croplands without cover crops compared with annual croplands with cover and those under perennial production (Fig. 5B). Furthermore, perennial-based systems had significantly higher SEMWISE ratings than annual cropland with cover cropping (Fig. 5B). These results support the important role of adding a cover crops (Blanco-Canqui et al., 2015) or perennials (Chamberlain et al., 2022) to improve overall soil health. In contrast, CASH did not show a difference between annual cropland with cover crops and perennial-based systems with cover crops and rangeland (Fig. 5A). The adjustments for climate and soil texture using SEMWISE thus improved the sensitivity of the model and the ability to detect significant differences between management groups.

We wanted a robust tool that not only can detect broad differences at a large, national scale, but also at finer resolutions, such as those related to research at a local scale. Samples collected from a long-term agroecosystem trial in California provided this opportunity ($n = 21$). Among the multiple trials at this site, plots were established to evaluate the effect of cover crop and N fertilizer type (e.g., organic vs. synthetic) in a corn-tomato rotation on soil and crop productivity metrics. The median SEMWISE rating was 9-times greater in the synthetic N rotation with a cover crop compared to the same system without a cover crop (Fig. 6). Additionally, the SEMWISE soil health rating was 1.4-times greater in

the cover crop plots with organic N fertilizer compared to those with synthetic N, illustrating the ability of SEMWISE to detect responses to management using relatively few samples. Thus, this framework can be applicable to a wide variety of purposes.

3.3. How well does the soil health management index (SHMI) align with SEMWISE scores?

The construct of soil health emphasizes that soil is a complex, living ecosystem and as such, management practices evaluated in isolation of other components of the system may underscore the overall effect of how a soil functions. Ideally, a soil health management system will be designed to address all four of the soil health principles – minimally disturb the soil, protect the soil with cover, optimize biodiversity, and maintain a living root for as long as possible. Evaluating soil health based on the integrated effect of these principles has proven challenging and typically has been done by evaluating only one or two practices at a time (Horwath, 2022). No metrics currently exist to translate and quantify management practices in terms of the principles. The Soil Health Management Index (SHMI), built from the combined influence of the soil health principles, provides an initial framework to address these challenges at a system-level.

Each of the three indices representing key soil health principles were designed to best utilize the existing metadata provided in this survey. Four bins were created to group the samples into relatively equal sample sizes (Table S7) and to put the quantitative (richness and roots) and categorical (disturbance) indices on the same scale. The bins ranged from bin 1, representing the “lowest” soil health conditions, to bin 4, representing the “highest” soil health conditions. Among the management indices developed, we obtained comprehensive information on crops planted over a 2–3-year period enabling us to calculate a plant richness index. Richness bin 1 included samples with less than one average annual functional richness category. This tended to be the case when fallow was included in the crop rotation in low-plant diversity systems (e.g., a wheat-fallow system). Bin 2 included samples with one category, common in monocultures such as continuous corn or cotton. Bin 3 included samples greater than one and less than or equal to three categories. A classic example for bin 3 is a corn-soybean rotation with a rye cover crop. Bin 4 was the most diverse system and included all rangeland sites and annual cropland that included a diverse mixture of cover crop species, and it included samples with more than three

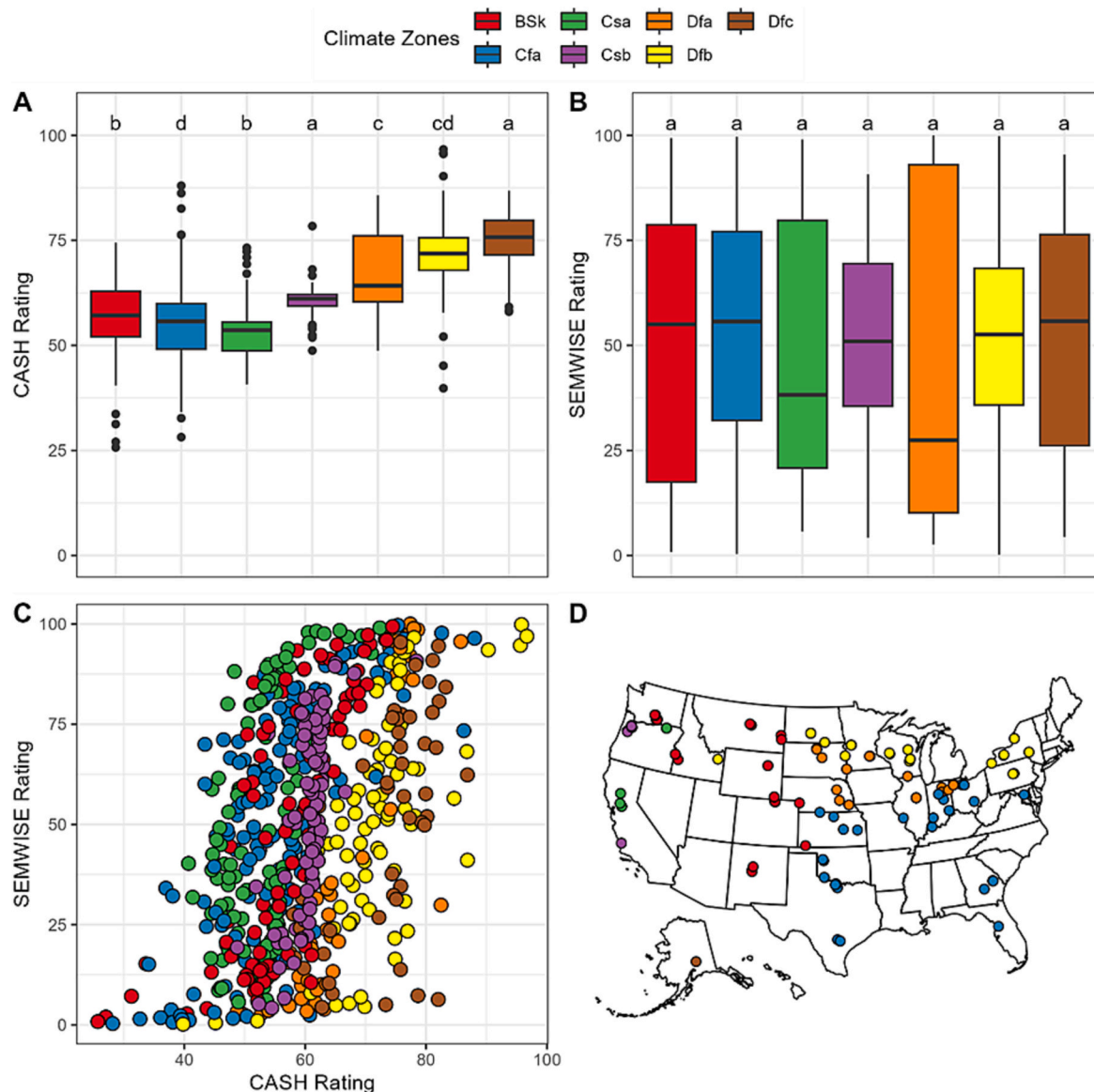


Fig. 4. CASH and SEMWISE ratings by Köppen climate zone. A) Distribution of CASH ratings by climate zone, B) distribution of SEMWISE ratings by climate zone, C) scatterplot comparing the CASH and SEMWISE ratings, D) the sampling locations represented by this data set. Statistical comparisons were made using a Wilcoxon rank sum test (significance threshold of $p < 0.05$). Since the SEMWISE rating does not include nutrient data, similar comparisons were made using a CASH rating without nutrients, and results were similar (data not shown).

categories.

Initial efforts to collect information regarding disturbance were focused on two broad categories: 'conventional till' and 'no-till'. Despite follow-up requests for additional information about frequency or depth, responses with additional details were too limited to expand beyond tillage type. For the disturbance index, bin 1 included conventional till, disc plow, and moldboard plow, bin 2 included chisel plow, cutting disc, shallow till, strip till, vertical till, and zone till, bin 3 included no till and no till/direct seed, and bin 4 included samples in the "none" category. Future SHMI scores would benefit from additional disturbance information such as specific tillage type, tillage frequency, and depth, etc. Utilization of tools such as the NRCS Soil Tillage Intensity Rating, which is a measure of soil disturbance that reflects all field operations, not just cultivation, would be an ideal candidate for this purpose.

Our roots index assumed that the soil health principle of cover was adequately captured because information on the amount of soil coverage by growing plants or mulches was not included in the meta-data. Better cover data could be acquired directly using satellite imagery

(ideally) or indirectly through planting data such as planting density and row spacing as well as residue management practices following harvest. For the roots index, bin 1 had roots present less than or equal to 50 % of the time, bin 2 had >50 % and less than or equal to 67 % root presence, bin 3 had >67 % and <100 %, and bin 4 had roots present 100 % of the time. Improvements to the root index calculation could be achieved by acquiring actual planting, harvest, and termination (for cover crop) dates (we used state averages, provided from USDA-NASS; see [Section 2.5](#) Calculating soil health management index and connecting SEMWISE to management) and possibly adjusting for a minimum growth period beyond germination based on growing degree days and a target physiological state.

By comparing SHMI and SEMWISE, we sought to determine whether SEMWISE could detect management at a systems level; SHMI scores provide an overall system management index that is based on the soil health principles. In general, the SEMWISE rating increased with each successive bin of our three soil health principle indices ([Fig. 7](#)). Rangeland systems ranked the highest for each index and had the highest

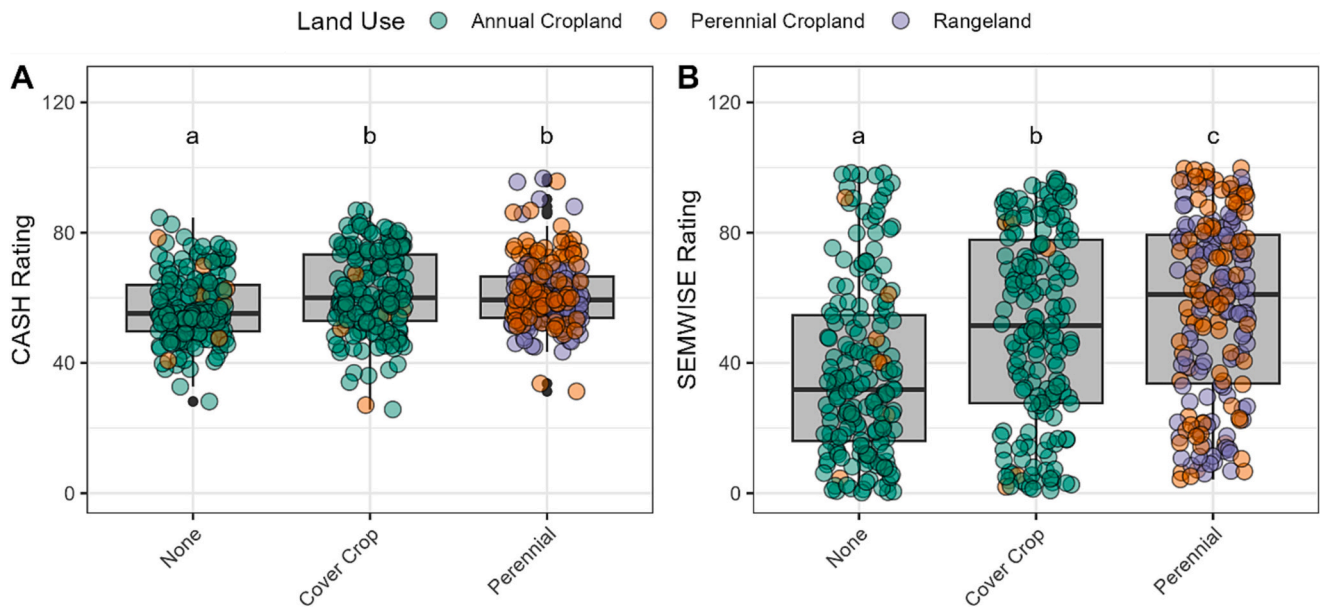


Fig. 5. CASH and SEMWISE ratings by cover type. Points represent the individual values for each soil sample and are colored by annual cropland (green), perennial cropland (orange), and rangeland (purple). Statistical comparisons were made using a Wilcoxon rank sum test (significance threshold of $p < 0.05$). Since the SEMWISE rating does not include nutrient data, similar comparisons were made using a CASH rating without nutrients, and results were the same (data not shown). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

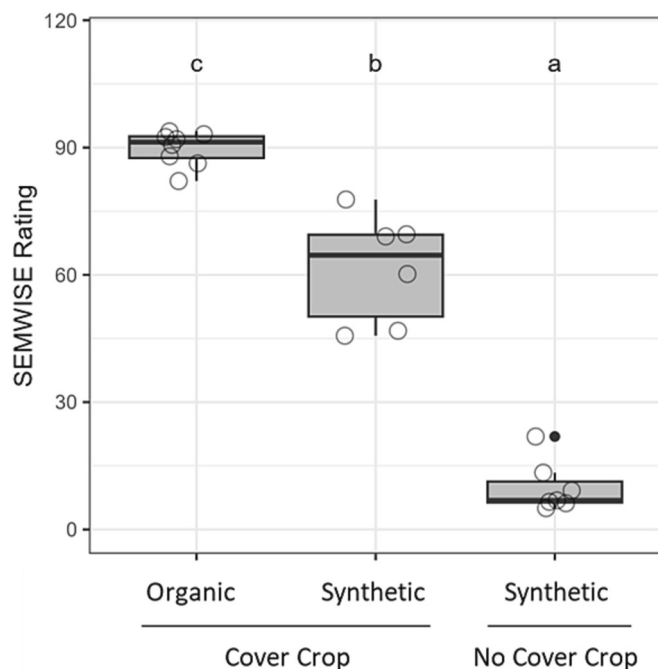


Fig. 6. SEMWISE ratings show the effect of cover crop and N fertilizer source in a corn-tomato rotation in California. Statistical comparisons were made using a Wilcoxon rank sum test (significance threshold of $p < 0.05$).

average SEMWISE soil health scores, further supporting enhanced soil health with diverse vegetation, continuous living roots, and minimal disturbance provided sound grazing management is implemented. Within cropland sites, improved soil health was associated when roots were present for at least two-thirds of the year, two to three plant functional groups occurred in the rotation, and either no-till or conservation tillage practices were used (Fig. 7). The relatively high SEMWISE ranking of disturbance bin 2 (conservation tillage) compared to bin 3 (no-till) is likely attributed to most samples associated with perennial-

based systems (e.g., vineyards, hop yards, orchards, etc.), where disturbances tend to be less frequent and shallower than in annual crops (samples were collected within the crop row and not in alleys). Although soil health has been linked to individual management practices (e.g., Horwath, 2022), this is the first known report showing a strong link between soil health indicators and a parameterized index of soil health principles that spans a large array of agricultural systems differing in crop rotations, plant biodiversity, and tillage practices across the U.S. Although our focus was to test a singular SEMWISE score to the various soil health principles, the SEMWISE framework also allows users to calculate and evaluate ratings associated with each individual indicator (Figs. S3–S9).

Agricultural management is the synthesis and integration of all soil health principles, which may or may not be mutually exclusive on any given site. For example, increasing plant richness by adding a cover crop will increase both the roots and richness scores. It is also true that while each soil health principle is important for overall soil health, each principle may have overlapping benefits. For example, both plant biodiversity and root presence contribute to the quantity and diversity of organic matter and root exudates that is necessary to feed and support a healthy soil microbial community (Adamczyk et al., 2019; Eisenhauer et al., 2017; Lange et al., 2015). It is also true that some principles may not be available or feasible in all cropping systems (e.g., no-till in potato). In the potato example, incorporating perennials into the rotation or adding a diverse cover crop could mitigate the necessary soil disturbance through harvesting root crops (Powell et al., 2020). We contend that there are multiple avenues for the production system to achieve high soil health and thus created a single management index to better describe the overall potential of an agricultural system. This combined soil health management index (SHMI) is the sum of the roots, richness, and disturbance bins and represents the collective management at the system level. The five SHMI bins were calculated by summing the three indices and grouping results into multiples of two (bin 1 = 3 or 4, bin 2 = 5 or 6, bin 3 = 7 or 8, bin 4 = 9 or 10, bin 5 = 11 or 12).

Our overall SHMI score also increased with our SEMWISE soil health scores that were relevant to the management systems in our national study (Fig. 8). The highest SEMWISE ratings were associated with the highest SHMI bin (5), which was represented primarily by rangeland

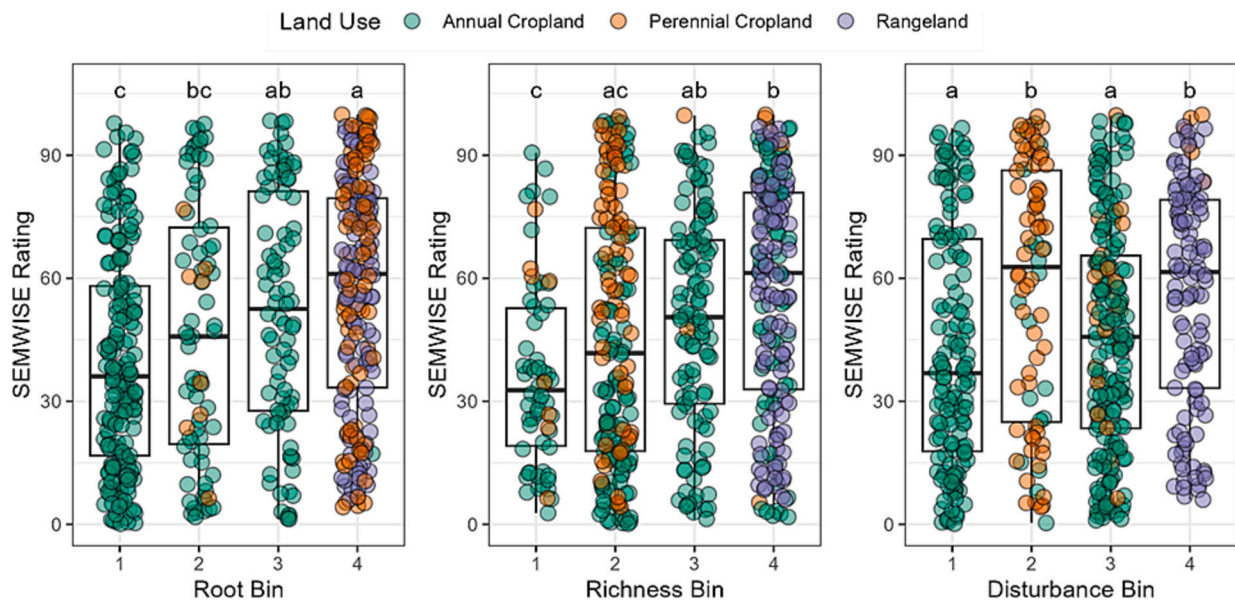


Fig. 7. SEMWISE ratings across root, richness, and disturbance bins. Points represent the individual values for each soil sample and are colored by annual cropland (green), perennial cropland (orange), and rangeland (purple). Statistical comparisons were made using a Wilcoxon rank sum test (significance threshold of $p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

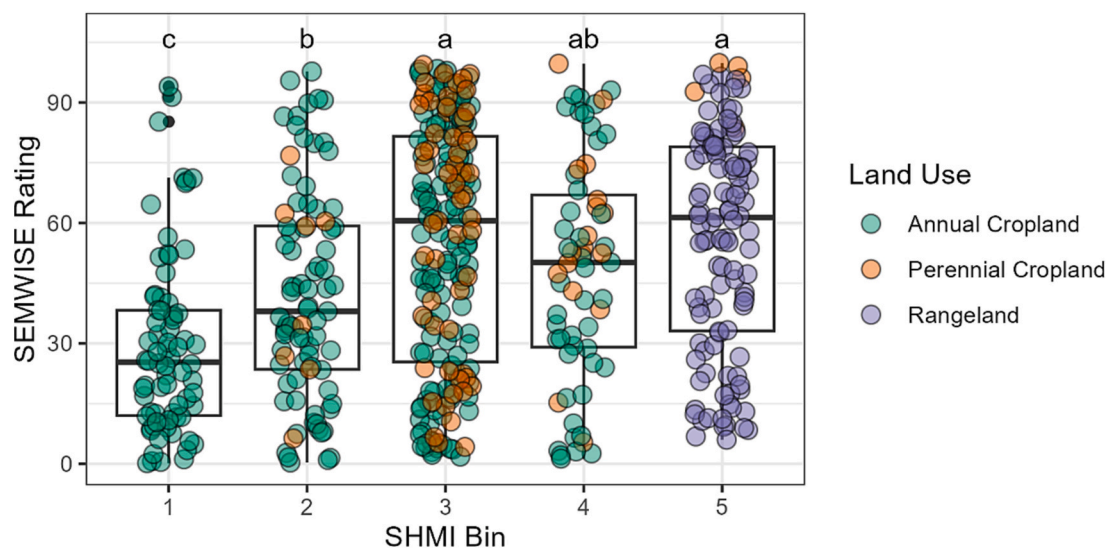


Fig. 8. SEMWISE ratings across Soil Health Management Index (SHMI) bins. Points represent the individual values for each soil sample and are colored by annual cropland (green), perennial cropland (orange), and rangeland (purple). Statistical comparisons were made using a Wilcoxon rank sum test (significance threshold of $p < 0.05$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

samples with high plant diversity (rating of four for richness), continuous roots (and cover) (rating of four for roots), and the least soil disturbance (rating of four for disturbance). Samples in SHMI bin 5 were significantly greater than all other bins except bin 3. Samples in bin 3 were comprised primarily of annual cropland that was not tilled or conventionally tilled and had a diverse cover crop or diverse crop rotation, or there was a perennial crop in the rotation. Bin 4 was an anomaly in the data set as these samples were characterized as either perennial based or cropland with high diversity in either the cover crop when present or in the crop rotation. It is unclear why these samples had a lower median SEMWISE score than bin 3 or bin 5, but it may have been related to the short time under which the systems these samples were collected from were practicing soil health principles (personal communication with sample collectors). Further improvements to each index and the overall SHMI should include the duration each practice has been

in place since the response of many indicators may take multiple years to be realized (Moebius-Clune et al., 2016; Stott, 2019). Samples in bin 2 represented annual cropland that were conventionally tilled with a cover crop and those that were no-till without a cover crop and had a fallow period in the rotation. The samples with the lowest SEMWISE score also had the lowest SHMI rating (bin 1) and were all characterized by conventional tilled cropland with many plots containing fallow in the rotation. The strong correlation between SEMWISE ratings and SHMI validates the ability of this framework to interpret the soil health, even in the presence of complex management systems.

4. Limitations and conclusions

There are some limitations in this study that should be addressed in future work. There were data shortages on several components of

disturbance that could have been used to develop a more informed disturbance index, including the type of tillage instrument, depth of tillage, wheel speed of the tractor, slope gradient, and direction of the tillage operation. Additionally, we did not have exact planting and harvest dates for the root index. Besides qPCR, this study only used soil health indicators that were available through the Cornell Soil Health Laboratory, and it is possible that other soil health indicators could be useful in the SEMWISE model. While this study was inclusive across many climates in the United States, SEMWISE should not be used to score soil health in other regions until the model includes samples from those areas.

Despite these limitations, our study was robust and inclusive enough to investigate many interesting questions. We aimed to answer the question if it was possible to design a tool that could facilitate comparisons of soil health indicators at a national scale. Our new framework, SEMWISE, used structural equation modeling with indicators known to respond to management for a well-informed evaluation of soil health. SEMWISE was designed to adjust soil health indicator values to simultaneously account for differences in clay content and climate, thereby allowing for comparisons at multiple scales. The resulting SEMWISE model contained a reduced dataset of soil health indicators and ratings that could be compared individually or collectively in a SEMWISE score. Furthermore, we developed a second framework that integrated individual practices into a system-level SHMI, reflecting the principles of soil health to provide continuous roots and soil cover and a diverse plant composition, and to minimize physical disturbances to the system. SEMWISE scores significantly associated with increasing SHMI scores, supporting our overall goal to evaluate soil health at a systems level with the added functionality of decoupling individual indicators or principles for further interpretative power. Additionally, the SEMWISE and SHMI frameworks were robust, sensitive, and scalable for detection of soil health differences at national and local levels. Several potential improvements to the model were outlined and included the collection of additional metadata to enhance the resolution of SHMI. Future tools could use the framework described here to test alternative indicators and other underlying factors to interpret outcome-based assessments. Further development of this tool would allow for consistent interpretations of soil health and improved communication between researchers, producers, conservationists, and other stakeholders.

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CRediT authorship contribution statement

Heather L. Deel: Data curation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Jennifer M. Moore:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Supervision, Visualization, Writing – original draft, Writing – review & editing. **Daniel K. Manter:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare they have no conflict of interest.

Data availability

The authors do not have permission to share data, but all code used with the data is available on https://github.com/DanielManter-USDA/SEMWISE_SHMI.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsoil.2024.105273>.

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